

The Development of a Common Factors Based Virtual Reality Therapy System for Remote Psychotherapy Applications

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ABSTRACT

In person psychotherapy can be inaccessible to many, particularly isolated populations. Remote psychotherapy has been proposed as a more accessible alternative. However, certain limitations in the current solutions including providing a restorative therapeutic environment and therapeutic alliance have meant that many other people are left behind and do not receive adequate treatment. A common factors based VR and EEG remote psychotherapy system can make remote psychotherapy more accessible and effective for people in which current options are not sufficient.

Keywords: Virtual Reality, EEG, Remote Psychotherapy, Common Factors Theory

1 INTRODUCTION

Mental health disorders such as depression and anxiety affect one in five adults in the United States. According to the 2019 National Survey on Drug Use and Health (NSDUH), non-serious mental illnesses are found in 29.4% of young adults aged 18-25 years old and 25% of adults aged 26-49 years old [1]. A common method of treating these conditions is counseling that is directly by a therapist or psychiatrist. In 2019, the NSDUH found that only 44.8% of all adults living with non-serious mental illnesses sought treatment. However, several barriers prevent patients from seeking treatment. These barriers include the stigmatization of mental illness, confidentiality and trust, difficulty identifying the symptoms of mental illness, lack of accessibility, and fear or stress about the very idea of seeking treatment for mental illness [2] [3] [4] [5] [6]. In 2020 and 2021 with the rise of the COVID-19 pandemic, 41.5% of US adults reported to have been struggling symptoms of an anxiety or depressive disorder [7]. With this added burden, the increase in social isolation during the pandemic, and unknown long term psychological effects of the past year, the need for an effective remote psychotherapy treatment is even more evident.

Isolated populations

Isolated populations can refer to people who are geographically, psychologically, linguistically, or economically isolated. In psychotherapy, geographically isolated populations are populations where the nearest available therapist is too far away

for in person therapy to be a viable option for treatment. treatment. For example, in some areas of western Kansas the nearest therapist office could three hours away by driving. To ask people to travel three hours or more every week there and back provides an immense barrier to treatment.

Psychologically isolated populations refer to people who may find the idea of getting out of bed in the morning, getting dressed, leaving their homes, interacting with other humans, and talking to a therapist, impossible. In this context, economically isolated refer to populations where the cost of in person therapy is prohibitive [8] [9] [10] [11] [12] [13].

Current Remote Therapy Options

Currently, several remote psychotherapy options are available on the market in various forms of telehealth including video chat, text based, and phone call based therapy. Some examples of these treatments include BetterHelp, Talkspace, Zoom chat based therapy, online resources, educational seminars, and others. These resources have provided a valuable lifeline for many people during this difficult time.

However, both therapists and clients have had to adjust to this new paradigm of treatment. When therapists are no longer in the room with their clients, it becomes more difficult to recognize nonverbal cues and form connections with their clients [14] [15]. Likewise, since clients may be stuck at home or socially isolated, the environments for which treatment occurs may not be conducive to healing. For example, consider a client who schedules an appointment with their therapist for a session to work through stress due to working from home. During the session, they see work papers, an overloaded calendar, visible due dates, and a packed to-do list all scattered around the room and in sight from where they sit in front of their computer screen. It becomes difficult to focus on healing with all these distractions present. Furthermore, isolated populations are particularly vulnerable to these struggles irrespective of the pandemic. For psychologically isolated populations, the prospect of opening a Zoom chat staring face to face directly with a therapist can cause too much anxiety for them. At the same time, with no visual of the therapist, it can become too impersonal. Finally, for the clients that struggle to even get out of bed, they ultimately must receive treatment in the same space in which the worst of their struggles have occurred [14] [15].

2 THEORY: COMMON FACTORS APPROACH

Common factors theory refers to the psychological theory that several common factors need to be present for effective therapeutic healing to occur. These factors include empathy, alliance, environment, expectations, positive affirmation, genuineness, and cultural effects [16] [17] [18] [19] [20] [21]. For the purposes of this research, we will focus on the therapeutic alliance and the

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therapeutic environment, because these are the two factors in which current remote therapy options struggle with the most.

The therapeutic alliance refers to the ability for the therapist and their client to build a working relationship with each other. This requires the therapist to build trust with their client and form a partnership, so that the client can feel like they can work together with their therapist towards healing. In remote psychotherapy, it can be very challenging to adequately foster the therapeutic alliance. Therapists have several techniques to quickly build a relationship with their clients including mimicking body language and energy, building rapport, and probing their clients at the right moments. All three of these techniques are much more challenging remotely [22] [23].

The therapeutic environment refers to external surrounding environment in which the therapy takes place. For healing to occur, the client must be in a space where they feel comfortable, safe, and open to being vulnerable. Examples of key features of a restorative space include having plants present in the room, having large windows so that the outside world is visible, and having comfortable seating. Nature therapy takes this idea further by taking their clients out into the natural environment. Nature has been associated with healing for centuries in various cultures.

2.1 Virtual Reality Based Psychotherapy

Virtual reality (VR) has been proposed as a potential solution to the limitations of current remote psychotherapy methods, because of its heightened sense of presence and emotional response in comparison to other forms of media. VR has been used successfully in the treatment of post-traumatic stress disorder (PTSD), various phobias, and anxieties. It has also been used to help foster better self-body image, self-compassion, and compassion in others. VR is also a valuable tool in psychotherapy, because of its 360°, immersive characteristics. VR can “transport” the client into a new environment of their choosing. Thus, therapists have control over their client’s environment, and even though their physical environment may not be conducive to therapy, their virtual environment can be. Be it space full of traumatic memories, a room full of stressful reminders, or a cramped space on the road, VR offers an escape from that [24] [25] [26] [27].

We propose in this research that VR environments designed specifically with the characteristics of restorative environments can help increase the effectiveness of remote psychotherapy by providing clients with a suitable therapeutic environment during therapy sessions.

2.2 Fostering the Therapeutic Alliance

The therapeutic alliance is the ability for therapists and patients to build to working relationship with each other built on trust. Therapists are trained to foster this relationship by initially building rapport with their patients, communicating clearly about goals for the sessions, and probing their patients at the right moments. To build rapport quickly, therapists are taught to mimic their patients body language and pick up on non-verbal cues. With much of the non-verbal communication lost over distance, therapists are unable to understand how their patients are feeling during the sessions, and building rapport becomes more difficult. Thus, it becomes more challenging for them to build trust. Additionally, therapists must push their patients outside of their comfort zone at the right moments to help facilitate positive behavioral change. To identify the right moments to do so, therapists must be able to gauge how their patients are feeling in real time. With experience,

therapists can become very adept at this skill in in-person therapy. However, in remote psychotherapy through video chat, phone, text based therapy it is increasingly difficult [22] [23].

Electroencephalography (EEG) has been used extensively to identify people’s emotions in real time including identifying people’s focus during gaming, students’ engagement levels, and people’s anxiety levels in a variety of applications. It has also been found that a single frontal (FP1) electrode EEG is sufficient in identifying these basic emotions. In EEG, brain activity levels and thus emotions are associated with the frequency levels of the signal. Alpha (8 to 12 Hz) is associated with relaxation, Beta (12 – 30 Hz) is associated with alertness, Gamma (30+ Hz) is associated with creativity and flow, Delta (0.1 – 3 Hz) is associated with dreamless sleep, and Theta (4 – 7 Hz) is associated with dreaming. Different combinations of these frequency bands have been associated with other base emotions including engagement, attention, and anxiety. It is proposed that the real time display of emotions during therapy sessions using EEG can help facilitate therapists’ ability to make connections with their patients and thus can help facilitate the therapeutic alliance [28] [29] [30].

2.3 Tracking the Treatment

Beyond the real time benefits during therapy sessions, neurofeedback can be useful for therapist to track their treatment and the progress of their patients. With the data collected from EEG, therapists will be able to see how their patients are feeling in their sessions, and over time they will be able to see trends of arousal, engagement, and attention throughout. This way therapists will be able to track how well their patients are progressing in their treatments. It can also be useful to see which techniques the patients respond to the most, and thus, therapists can adjust accordingly.

3 DESIGN AND DEVELOPMENT OF VR THERAPY SYSTEM

As stated in the introduction section, the purpose of this design approach was to create a virtual reality based therapy system for remote psychotherapy applications that adequately addressed the common factors of therapeutic healing. Specifically, the goal of this design was to create a therapy system that had an environment more conducive to therapy and more effectively facilitated the therapeutic alliance. To understand the challenges and limitations of remote psychotherapy, our first action was to interview several counselors across the rural areas of western Kansas. We found many therapists longing for help and a new approach to reach a population that either was unable to receive treatment or resistant to it. One therapist described a situation in which her clients would be required to travel four or more hours to and from the therapist office each week for treatment. Thus, she would frequently experience client drop off.

In addition to interviewing practicing counselors, key design elements were chosen based on the established literature relating to common factors theory, restorative environments and fostering the therapeutic alliance, as well as increasing presence in VR. The key elements of the therapy system include the Virtual Reality Environments, Therapist Avatars, EEG neurofeedback system, Therapist and Patient Interface, and the Overarching Remote Therapy Elements.

3.1 Design of Virtual Reality Environments

The key tenets that informed the design decisions as it relates to the VR environments came from research on restorative environments and presence in VR. An example of one of the key elements of both

building restorative environments and increasing presence in VR is incorporating agency for the client during the therapy session. To increase agency, two key design decisions were made: giving the client options to choose between different environments and giving the client the ability to move around the environment freely. Thus, the overall VR design was based on an open world concept, and I developed three different VR environments to choose from [31] [32] [33] [34]. Each environment will be discussed and its connection to specific elements of restorative environments will be discussed in more detail below. The virtual reality environments were developed in Unity3D. Assets included in the environments were designed in Autodesk Maya and textured in Adobe Photoshop or were taken from free open source packages included in Unity3D.

3.1.1 Forest World

The first and most fully developed VR environment included in the novel therapy system is the forest world environment. This environment is intended to mimic the restorative qualities of the natural environment and nature therapy. The goal of nature therapy is for the counselor and client to take a journey together through the natural environment. They will walk, hike, jog, or explore together all while conversing together in a therapy session. Additionally, whenever the client feels comfortable, they will pick a place to stop, soak in the environment, relax, and have deeper conversation. In the creation of the natural environment, we had to be mindful of several things. We choose a forest world because of the associated sense of growth and new beginning in plants and greenery. We incorporated ponds and a waterfall, because of the healing effects of water and the soothing sounds of rushing water. Finally, other design elements including the archway of trees, the variation in terrain elevation, incorporated clearings and nooks, and the backdrop of a mountain were intended to give clients interesting places to explore without being overly distracting [31] [32] [33] [34].

Overall, the intended use of this VR environment would be for the counselor and client to start the session by exploring spaces in the environment, allowing for the client select a comfortable place for them to stop, and finally conduct the therapy session side by side in the comforting environment of the client's choosing. **Figure 1** depicts the various design elements included in the develop VR forest world.



Figure 1. Images of the entire VR forest world environment. Design elements visible in this image include, the forest and pathway, ponds and waterfall, and the mountain backdrop.

3.1.2 Log Cabin

The second VR environment developed for the therapy system is the Log Cabin. This environment was developed to act as an indoor alternative to the forest world that follows the principles of

restorative environments. These principles include large windows with access to nature, soft soothing colors, large comfortable couches, and a crackling fireplace [34].



Figure 2. An image of the log cabin environment meant as an indoor alternative to the forest world.

3.1.3 Freud Therapist Office

The third virtual environment is the Freud Therapist Office, which is inspired by Sigmund Freud's therapist office in Vienna, Austria. Characteristics of this environment including a large comfortable sofa, several books, dark colors, ornate designs, and walls covered in pictures of psychology and neuroscience related images. One addition to the environment of note is that the client is placed lying on their back on the sofa to mimic the classic description of Sigmund Freud's therapy sessions [34] [35].

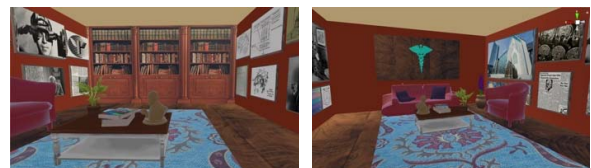


Figure 3. Images of the Sigmund Freud therapist office.

3.2 Therapist Avatars

Another important aspect of the therapy system is the inclusion of therapist avatars. The designs of the avatars are based on Jungian archetypes for therapeutic healing and created by graduate student and artist Kaylie Lyons. Each avatar is designed as a visual representation of one of the four key therapist characteristics that clients value: 1. Expertness and Wisdom, 2. Compassion and Empathy, 3. Similarity and Like, 4. Genuineness and Trustworthiness. Already in development are two avatars: a sage and a healer. The sage is meant to represent a wise old man and is intended for clients who value expertness and wisdom. The healer is meant to represent a woman Madonna figure and is intended for clients who value compassion and empathy. Future avatars will include characters that address the other key therapist characteristics. The virtual avatars will be developed using Autodesk Maya and textured using Adobe Photoshop. Currently, the healer is fully developed and included in the therapy system. At this point in time, the healer will be used as the avatar in the following studies [36] [37].



Figure 4. The designs for the healer and sage by Kaylie Lyons.

3.3 EEG Neurofeedback System

The EEG neurofeedback system is proposed as a potential solution for the difficulties of fostering the therapeutic alliance in remote psychotherapy. The neurofeedback system will measure brainwaves using the commercially available headset Neurosky Mindwave Mobile 2 Headset. Code for the real time signal processing, user interface, and display were developed in MATLAB using the Neurosky research tools package for MATLAB. The real time display of EEG is adjustable for therapists, where they will be able to choose to display any combination of the 1. raw signal, 2. frequency gradient, 3. Alpha waves, 4. Beta waves, 5. Gamma waves, 6. Delta waves, 7. Attention data, and 8. Meditation data. Alpha (8 to 12 Hz) is associated with relaxation, Beta (12 – 30 Hz) with alertness, Gamma (30+ Hz) with creativity and flow, Delta (0.1 – 3 Hz) with dreamless sleep, and Theta (4 – 7 Hz) with dreaming [28] [29] [30].

3.4 Therapist and Patient Interface

An important aspect to an effective remote therapy system is an approachable and useful therapist and patient interface. For the patients, the interface must be streamlined, easy to use, and customizable. It is also critical for patients to feel like that have control over the experience. This will help to add agency for the patients and can increase their comfort level with using the therapy system. Designs for the patient interface were developed using Adobe XD as a potential template for what a mobile app of the system would look like. Critical elements of the patient interface include the ability to choose a therapist based on their pictures and individual bios, the ability to choose their desired avatar, and the ability to choose their virtual environment of preference.

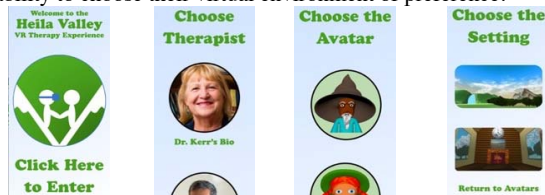


Figure 5. Design of patient interface of therapy system developed in Adobe XD.

The designs are based on interviews with four therapists about what the most important elements a remote therapy interface should include. These elements included a space for clients and their clients' profiles (profile picture, demographic information, overall goal in seeking therapy, consent forms, recording of EEG data and progress, and notes from sessions), live EEG signal display, calendar, inbox, and a space for notes. The EEG display can be customized to show whatever signals therapists deem most critical and can be clicked into to view in detail. The design for the therapist interface was created in Adobe XD. Other important design criteria to note in the future include privacy concerns, which could be addressed by using a doubly secured webpage and a data encrypted signal connection.

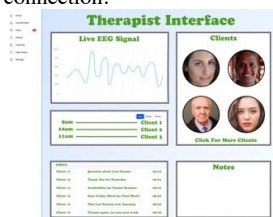


Figure 6. Design of therapist interface of therapy system developed in Adobe XD.

3.5 Overall Therapy System

In addition to the virtual environments, avatars, EEG neurofeedback system, therapist interface, and patient interface, an effective remote psychotherapy system must securely connect therapist and patient over distance and seamlessly combine all the elements of the remote therapy system. For use in this study, the full therapy system will include the VR mobile headset, EEG headset, developed VR iPhone mobile app, live data processing of EEG in MATLAB, connection between therapist and patient over Zoom, and live stream of the patient view in VR using Teamviewer. Thus, the therapist communicates with the patient via Zoom audio, while being able to see the EEG data in addition to the image that the patient is seeing in VR. The patient, however, is immersed in VR with both VR and EEG headsets and is communicating with the therapist via Zoom audio. In practice, an online secure platform that combines all these elements would need to be developed.

4 FUTURE DEVELOPMENTS

Currently, we are in Phase 2 of the development of the common factors based therapy system. Phase 1 included all the conceptualization, design, and development outlined in this report. Phase 2 is the evaluation of the therapy system with counselors and clients. Four metrics will be used in two human subject studies, one focused on the functionality and usability of the system for counselors and another focused on the functionality and usability of the system for clients. The four study metrics are as follows: 1. Client Reactions Systems (CRS), 2. Perceived Restorative Scale (PRS), 3. Session Evaluation Questionnaire (SEQ), and 4. Presence Questionnaire (PQ). The purpose of the CRS is to assess the basic client reactions to a therapeutic intervention. In this context, it is being used as a way for the subjects to observe and assess the effectiveness of the therapy session [38]. The purpose of the PRS is to provide a metric to assess restorativeness of a therapeutic technique, environment, or system [39]. The SEQ is a standard metric for analyzing the effectiveness of a therapy session or technique. It has been found to be correlated with the therapeutic alliance [23] [40]. The purpose of the PQ is to give an assessment of the effectiveness of the virtual reality technology use in this system. Ultimately, presence is crucial for the client to feel engaged and immersed into the environment and the therapy session [41]. At the conclusion of Phase 2, Phase 3 of the development will include revisions of the therapy system based on the results from Phase 2 and the further development of full functioning prototype.

5 CONCLUSION

Mental illnesses affect millions of people in the United States alone, and over 40% of people struggling with mental illnesses do so without any treatment at all. Several barriers prevent patients from seeking treatment including stigmatization of mental illness, confidentiality and trust, lack of accessibility, and fear or stress about the very idea of seeking treatment [1] [2] [3] [4] [5] [6]. To address these barriers remote psychotherapy solutions have been implemented including text, phone call, and video chat based therapy systems. Over the past year and a half during the COVID-19 pandemic where in person psychotherapy was impossible these solutions have provided immense support for many of the people in need. However, certain limitations in current remote psychotherapy solutions including the challenges in providing a therapeutic environment and in facilitating the therapeutic alliance has meant that many other people are left behind and do not receive adequate treatment. In this report, it is proposed that a VR and neurofeedback based remote psychotherapy system specifically designed based on common factors theory can make remote psychotherapy more accessible and effective for people in which current options are not sufficient.

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